Non-Markovian Memory-Induced Effects in Quantum Cosmology

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Outline

Motivation and Background

A Causal Memory Kernel and Fractional Quantum Cosmology

The Primordial Power Spectrum

Primordial Non-Gaussianity

Conclusions and Outlook

Historical Context

- 1959-63: ADM Formalism:- Arnowit, Deser and Misner developed the $3+1$ decomposition of General Relativity, splitting spacetime into three-dimensional spatial slices evolving in time, and reformulated GR in Hamiltonian form.
- 1960s: Wheeler & DeWitt: Developed quantum geometrodynamics, quantizing spacetime geometry and describing the universe by a wave functional $\Psi[h_{ij}]$. Applying the Hamiltonian constraint gives the Wheeler-DeWitt equation.
- 1980s-Today Kiefer and collaborators develop the semiclassical expansion of the WDW equation.

Quantum cosmology

- The description of the early universe sits at the interface of gravitation, quantum theory, and cosmology.
- The Wheeler-DeWitt (WDW) equation is obtained from canonically quantizing general relativity in minisuperspace.
- Solving it in a controlled semiclassical expansion recovers classical cosmology, then quantum mechanics, and finally computable quantum gravitational corrections.
- These corrections give a concrete observational window into quantum gravitational imprints which modify the primordial power spectrum

The Kiefer semiclassical expansion

Expanding order by order in the Planck mass squared, $S = m_P^2 S_0 + m_P^0 S_1 + m_P^{-2} S_2 + \ldots$, gives:

Order	**Physics**	**Result**
m_P^4, m_P^2	Classical background	Hamilton Jacobi eq. $\Leftrightarrow$ Friedmann eq.
m_P^0	Quantum Mechanics	emergent Schrödinger equation.
m_P^{-2}	Quantum gravitational corrections	small, calculable, local in time (standard).

The m_P^{-2} order is where quantum gravity first enters and, in the standard treatment, it does so *locally* in time.

The Wheeler DeWitt equation

Background variables ($\alpha = \ln(a/a_0)$, scalar field ϕ) combine into a minisuperspace coordinate q^A with metric $\mathcal{G}_{AB}$,

$$\frac{1}{2}\left\{-\frac{1}{m_P^2}\,\mathcal{G}^{AB}\frac{\partial^2}{\partial q^A \partial q^B} + m_P^2\,V(q^A) - \frac{\partial^2}{\partial \nu_k^2} + \omega_k^2(\eta)\,\nu_k^2\right\}\Psi_k = 0$$

This is the starting point for the semiclassical (WKB) expansion.

The WKB expansion

Write the wave function as a phase and expand order by order in the Planck mass squared:

$$\Psi_k = e^{iS(q^A,\nu_k)}, \qquad S = m_P^2 S_0 + m_P^0 S_1 + m_P^{-2} S_2 + \dots$$

- Derivatives that act on this exponential ansatz are of integer order and obey the chain and product rule, this is what lets the expansion separate cleanly, order by order.

Leading orders

Order m_P^4: S_0 does not depend on the perturbations.
Order m_P^2: S_0 obeys a Hamilton Jacobi equation, equivalent to the classical Friedmann equation,

$$\mathcal{G}^{AB}\,\partial_A S_0\,\partial_B S_0 + V(q^A) = 0.$$

Order m_P^0: defining an effective conformal time η from S_0

$$\frac{\partial}{\partial\eta} = \mathcal{G}_{AB}\frac{\partial S_0}{\partial q_A}\frac{\partial}{\partial q_B},$$

yielding a Schrödinger equation,

$$i\,\frac{\partial}{\partial\eta}\,\psi_k^{(0)} = \mathcal{H}_k\,\psi_k^{(0)}, \qquad \mathcal{H}_k = -\frac{1}{2}\partial_{\nu_k}^2 + \frac{1}{2}\omega_k^2(\eta)\,\nu_k^2.$$

Where quantum gravity enters: order m_P^{-2}

The first genuine quantum gravitational correction:

$$i\frac{\partial}{\partial\eta}\psi_k^{(1)} = \mathcal{H}_k\psi_k^{(1)} - \frac{\psi_k^{(1)}}{m_P^2\psi_k^{(0)}}\left[\frac{(\mathcal{H}_k)^2}{V}\psi_k^{(0)} + i\,\partial_\eta\left(\frac{\mathcal{H}_k}{V}\right)\psi_k^{(0)}\right]$$

In the standard Kiefer treatment, this correction is still *local* in η: the equation depends only on the instantaneous state ψ_k, not on its history.

A Gaussian ansatz

- Specialize to a de Sitter background (constant Hubble rate H_0).
- Scalar and tensor perturbations obey the same effective mode equation,

$$\omega_k^2(\eta) = k^2 - \frac{2}{\eta^2}.$$

- Impose the Bunch Davies vacuum at early times; a Gaussian ansatz for the wave function turns the problem into a Riccati equation for the Gaussian width $\Omega_k(\eta)$, solvable in closed form.

The known quantum gravitational correction

Solving the m_P^{-2} Schrödinger equation for the Gaussian width gives the quantum gravitationally corrected power spectrum:

$$\mathcal{P}_{S,T}^{(1)}(k) = \mathcal{P}_{S,T}^{(0)}(k)\frac{\mathfrak{Re}\Omega_{\mathbf{k}}^{(1)}}{\mathfrak{Re}\Omega_{\mathbf{k}}^{(0)}} = \mathcal{P}_{S,T}^{(0)}(k)\left[1 + 0.988\,\frac{H_0^2}{m_P^2}\left(\frac{k_0}{k}\right)^3\right]$$

- $P_S(k) \sim k^{-3}$: enhanced at small k
- This form is a hallmark of the standard QG effects.
- The correction to temperature ansiotropies C_l is peaked at low ℓ (large angular scales) and rapidly suppressed at high ℓ (small angular scales).

Why Memory?

- Locality may break down at high energies, motivating non-local theories of quantum gravity that improve the renormalizability of gravity.
- Gravitational memory phenomena motivate the search for analogous and potentially richer memory effects in quantum gravity.

Fractional calculus is a natural language for describing such memory effects: a derivative of non-integer order depends on the function's entire history.

Fractional Derivatives

There exist many types of fractional derivatives, two prominent ones are

- **Caputo:-**

$$ {}^{C}D_{\eta}^{\beta} f(\eta) = \frac{1}{\Gamma(n-\beta)} \int_0^{\eta} \frac{f^{(n)}(\eta')}{(\eta-\eta')^{\beta-n+1}}\, d\eta' $$

  where $n = \lfloor \beta \rfloor$.

- **Riemann-Liouville**

$$ D_{\eta}^{\beta} f(\eta) = \frac{1}{\Gamma(n-\beta)} \frac{d^n}{d\eta^n} \int_0^{\eta} \frac{f(\eta')}{(\eta-\eta')^{\beta-n+1}}\, d\eta' $$

  Unlike Caputo derivatives, Riemann-Liouville derivatives doesn't allow initial conditions to be specified in terms of integer-order derivatives of the function of interest.

The naive approach, and why it fails

The most direct idea: replace integer order derivatives in the Wheeler-DeWitt equation (or the Kiefer equations) with Caputo fractional derivatives of order β.

- The Kiefer construction depends crucially on integer order derivatives acting on $\Psi = e^{iS}$, obeying chain and product rules, this is what lets the m_P^2 expansion separate cleanly.
- Caputo derivatives do not satisfy the chain and product rules, they do not, for example produce a closed Schrödinger equation.

Lesson: nonlocality cannot be bolted on at the level of differential operators. It has to be built into the physics at a more fundamental level.

Build memory more fundamentally

A generic memory correction modifies the evolution of a function by adding an integral over the function's past, weighted by a memory kernel:

$$h'(x)\big|_{\text{modified}} = h'(x)\big|_{\text{standard}} + \int_a^x K(x-x')\,h(x')\,dx'$$

We introduce exactly this kind of term into the Wheeler DeWitt equation itself, at order m_P^{-2}, the same subleading order where the standard local QG correction already lives (keeping the leading Hamilton Jacobi and Schrödinger equations untouched),

$$\frac{1}{2}\left\{-\frac{1}{m_P^2}\mathcal{G}^{AB}\partial_A\partial_B + m_P^2 V - \partial_{\nu_k}^2 + \omega_k^2\nu_k^2\right\}\Psi_k = \frac{1}{m_P^2}\int K(\eta-\eta')\,\partial_{\eta'}\Psi(\eta')\,d\eta'$$

Where does memory begin?

- We work in the Vilenkin tunneling picture (the same reasoning applies to Hawking Hartle no-boundary), with nucleation set at $\eta_{\rm nuc} = 0$.
- Memory only accumulates once a classical spacetime history exists, not before nucleation where the universe is in a classically forbidden region.
- This also fixes a mathematical problem: letting the memory integral run back to $\eta \to -\infty$ would generally make it divergent. Starting at $\eta_{\rm nuc} = 0$ keeps it finite.

Fixing the kernel's form

1. It must be causal, the evolution at η depends only on $0 \leq \eta' < \eta$, never the future.
2. It must exhibit the correct short distance behavior near $\eta' = \eta$.

There is also a physical reason for the kernel to depend on the perturbation mode k: different wavelengths sample different parts of the cosmological history, so they need not remember it equally well. This motivates the memory kernel:

$$K(\eta - \eta') = -iA \frac{\psi_k^{(1)}(\eta)}{\psi_{k_0}^{(1)}(\eta)} \left[\frac{1}{\eta - \eta'}\right]_+$$

where $[\cdot]_+$ is the Hadamard finite part (plus) distribution.

The kernel is secretly a Caputo derivative

The Caputo derivative of fractional order $\beta = 1 - \epsilon$:

$$ {}^{C}D_{\eta}^{1-\epsilon} f(\eta) = \frac{1}{\Gamma(\epsilon)} \int_{0}^{\eta} \frac{f'(\eta')}{(\eta - \eta')^{1-\epsilon}} \, d\eta' $$

Expanding for small ϵ reproduces the ordinary derivative at order ϵ^0 and the plus distribution structure of our kernel, with some additional terms, at order ϵ^1. Substituting back into the modified Wheeler-DeWitt equation, the m_P^{-2} dynamics reorganizes into a Schrödinger equation with a Caputo time-derivative of order β:

$$ i\,\eta^{\beta-1}(1 - \gamma_E + \gamma_E \beta)\; {}^{C}D_{\eta}^{\beta}\, \psi_k^{(1)}(\eta) = \mathcal{H}_k \psi_k^{(1)} + \mathcal{O}(m_P^{-4}) $$

Fractional time evolution is not put in by hand, it emerges as the small-ϵ limit of the causal memory integral.

The effective fractional order β

$$\beta = 1 - \frac{A}{m_P^2} \frac{\psi_{\mathbf{k}}^{(1)}(\eta)}{\psi_{\mathbf{k}_0}^{(1)}(\eta)} \simeq 1 - \mathcal{O}(m_P^{-2})$$

- The fractional order departs from unity by an amount controlled by A/m_P^2, Planck-suppressed, as it must be, since memory enters at order m_P^{-2}.

The Gaussian ansatz, now history dependent

- Repeat the Gaussian ansatz trick, now for the memory corrected Schrödinger equation.
- The normalization N_k and Gaussian width Ω_k no longer obey purely local differential equations, they satisfy integro-differential equations whose value at η depends on their entire past.
- This nonlocality can still be handled perturbatively in powers of m_P^2, and the leading order memory contribution can be computed in closed form.

The memory corrected power spectrum

$$\mathcal{P}^{(1)}_{S,T}(k) = \mathcal{P}^{(0)}_{S,T}(k)\left[1 + 0.988\,\frac{H_0^2}{m_P^2}\left(\frac{k_0}{k}\right)^3 + \frac{11A}{24m_P^2}\left(\frac{8}{3} - 2\ln 2\right)\left(\frac{k}{k_0}\right)^{3/4}\right]$$

Two independent corrections now sit side by side: the standard k^{-3} correction, and a new $k^{3/4}$ memory induced QG correction. The $k^{3/4}$ term is blue-tilted, it grows with k, the opposite behavior from the standard correction.

Two corrections, opposite tilts

Standard (Kiefer & Krämer)

$$P_S(k) \sim k^{-3}$$

- Arises through physics that is local in time, and is a direct consequence of the standard semiclassical Wheeler-DeWitt expansion.
- Red-tilted: strongest at small k

Memory-induced (this work)

$$P_S(k) \sim k^{3/4}$$

- Arises through a causal memory kernel and physics that is non-local in time.
- Blue-tilted: strongest at large k

From power spectrum to Temperature Anisotropies

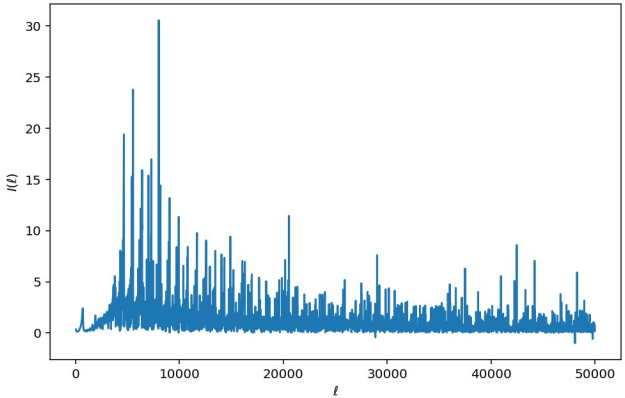
- The memory correction propagates through the CMB transfer function, sourcing a correction $\Delta[\ell(\ell+1)C_\ell/2\pi]$.
- **At low ℓ:** the memory term give a very small modification to the temperature anisotropy spectrum, large-scale anisotropies remain controlled by the standard spectrum.
- **At high ℓ:** this is where the blue-tilted $k^{3/4}$ correction becomes distinctive and potentially observable.

Evaluation of Undamped Temperature Anisotropies

- Numerically evaluating the undamped memory correction to

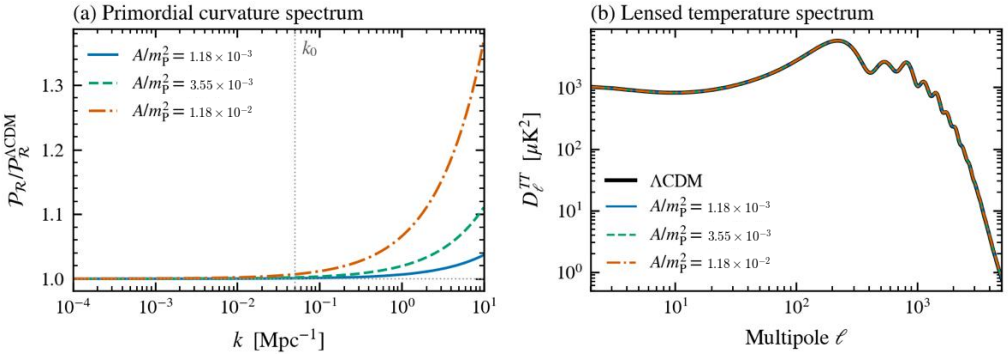
$$\Delta D_\ell = \frac{11 A_s A}{24 m_P^2 (k_0 \eta_{\text{rec}})^{3/4}} \frac{8 - 6\ln 2}{3} I_\ell.$$

- Memory effects are more pronounced at high ℓ, culminating in pronounced peaks of $I(l)$
- The extension of this undamped spectrum to $l \sim 5 \times 10^4$ is only intended to illustrate the behavior of $I(\ell)$. Damping effects aren't taken into account.



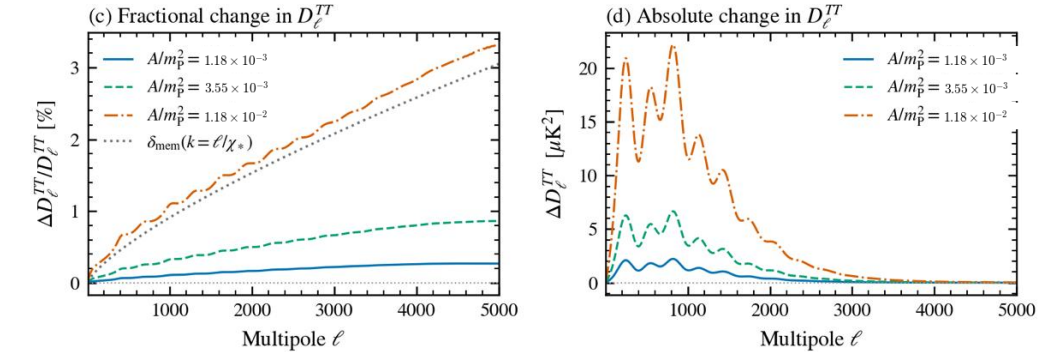
Numerical evaluation: $I(\ell)$ for $30 \leq \ell \leq 5 \times 10^4$.

A full Damped CAMB analysis (i)



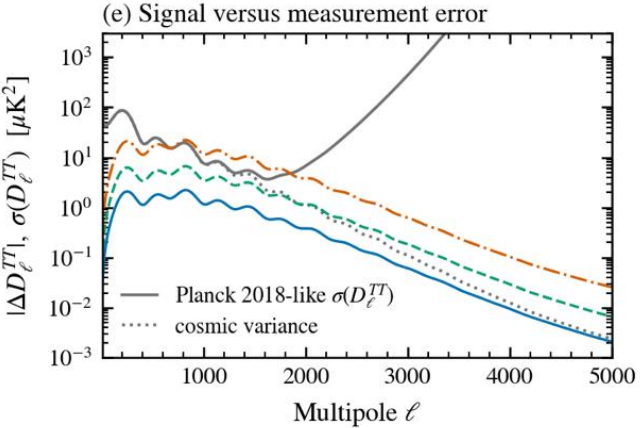
- **(a)** Ratio to ΛCDM: Very close to unity for $k \ll k_0$, but departs for $k >> k_0$, growing as $k^{3/4}$.
- **(b)** Lensed D_ℓ^{TT}, the spectrum stays visually similar with ΛCDM.

A full Damped CAMB analysis (ii)



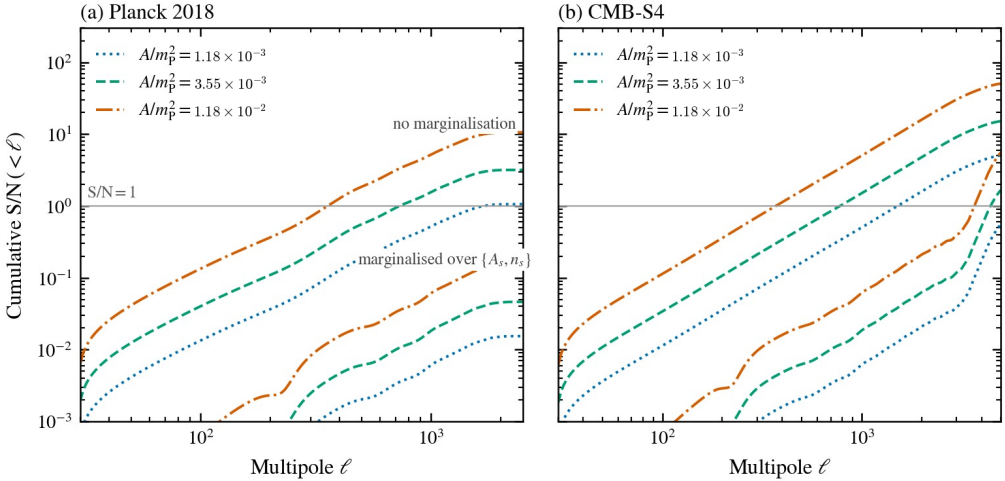
- **(c)** Fractional change keeps growing toward high ℓ, tracking the undamped $\ell^{3/4}$ scaling (grey dotted).
- **(d)** The absolute modification of the temperature anisotropy spectrum is suppressed by Silk damping past $\ell \simeq 5000$.

A full CAMB analysis (iii)



- **(e)** The signal exceeds cosmic variance over part of the range; for Planck-like data, instrumental sensitivity, not cosmic variance, limits detection at high ℓ.

Forecasting detectability: Planck 2018 vs. CMB-S4



Cumulative signal-to-noise on A/m_P^2 from $\ell_{\rm min} = 30$: (a) Planck 2018-like, (b) CMB-S4-like. Upper curves: parameters fixed. Lower curves: marginalised over $\{A_s, n_s\}$.

Next-generation experiments are sensitive to memory effects

CMB-S4-like sensitivity: marginalised cumulative S/N > 1

- **Unmarginalised:** for a Planck 2018-like experiment, the cumulative S/N already exceeds unity for large enough memory amplitudes.
- **But parameters are degenerate:** marginalising over the standard primordial parameters $\{A_s, n_s\}$, which can partly mimic a blue tilt, substantially reduces the Planck-like significance.
- **CMB-S4 changes this:** even *after* marginalising over $\{A_s, n_s\}$, the cumulative signal-to-noise exceeds unity for $A/m_P^2 \sim 10^{-2} - 10^{-3}$.
- **Take-home:** next-generation CMB experiments are sensitive non-Markovian memory effects.

Higher Order Correlation Functions

The same nonlocal memory kernel that modifies the two-point correlation function should generically also affect higher-order correlation functions. Write the memory-corrected power spectrum as $P_\zeta^{(1)}(k,\eta) = P_\zeta^{(0)}(k)[1+\delta_{\rm mem}(k,\eta)]$, with

$$\delta_{\rm mem}(k,\eta) = C_{\rm mem}\,\frac{A}{m_P^2}\left(\frac{k}{k_0}\right)^{3/4}, \qquad C_{\rm mem} = \frac{11}{24}\left(\frac{8}{3} - 2\ln 2\right) \simeq 0.587.$$

The squeezed limit

In the squeezed limit ($k_L << k_S$), the bispectrum is given by,

$$B_\zeta(k_L, k_S, k_S) \simeq \frac{12}{5} f_{\rm NL}^{\rm sq} P_\zeta(k_L) P_\zeta(k_S).$$

For standard single-field slow-roll inflation, we have the Maldacena consistency relation,

$$f_{\rm NL}^{\rm sq} = \frac{5}{12}(1 - n_s)$$

Memory-induced squeezed non-Gaussianity

$$\Delta f_{\rm NL}^{\rm sq} \simeq -\frac{5}{16}\,\delta_{\rm mem}(k_S) \;\simeq\; -0.183\,\frac{A}{m_P^2}\left(\frac{k_S}{k_0}\right)^{3/4}$$

- In the presence of memory, the usual squeezed-limit logic itself is only approximate, motivating a more general shape function $\mathcal{F}_{\rm sq}(k_L/k_S)$.

Equilateral configurations

For equilateral configurations, $k_1 = k_2 = k_3 = k$,

$$B_\zeta(k,k,k) = \frac{18}{5} f_{\rm NL}^{\rm eq} P_\zeta^2(k)$$

Simple external-leg rescaling of the mode functions gives only a small correction to $f_{\rm NL}^{\rm eq}$. The dominant correction instead comes from interactions of mode functions, which is now weighted by a kernel, producing a $k^{3/4}$ running together with a logarithmic enhancement $\ln(k/k_{\rm nuc})$.

$$f_{\rm NL}^{\rm eq,mem}(k) \sim \mathcal{C}_{\rm eq} \frac{A}{m_P^2} \left(\frac{k}{k_0}\right)^{3/4} \ln\left(\frac{k}{k_{\rm nuc}}\right),$$

Folded Configurations

For Folded configurations, $k_1 + k_2 \simeq k_3$.
The memory kernel can generate significant signatures in folded configurations because nonlocal-in-time interactions, generate combinations that become enhanced in folded limits. Standard single-field slow-roll inflation does not produce a sharply enhanced folded bispectrum, but memory naturally does:

$$B_\zeta^{\text{fold,mem}} \propto \frac{A}{m_P^2}\left(\frac{k}{k_0}\right)^{3/4} P_\zeta(k_1)P_\zeta(k_2)\,\ln\left|\frac{k_1+k_2+k_3}{k_1+k_2-k_3}\right| + \text{perms.}$$

This folded enhancement could be one of the cleanest non-Gaussian signatures of the memory paradigm.

The non-Gaussianity running

If the dominant memory contribution scales as $f_{\rm NL}^{\rm mem}(k) \propto k^{3/4}\ln(k/k_{\rm nuc})$, its logarithmic running works out to a strikingly clean number:

$$n_{\rm NG}^{\rm mem} = \frac{d\ln|f_{\rm NL}^{\rm mem}|}{d\ln k} \simeq \frac{3}{4} \approx 0.75$$

A strong, distinctive scale dependence, far from the nearly scale-independent non-Gaussianity of minimal slow-roll inflation.

How big could this Non Gausianity be?

Squeezed : $\delta_{\rm mem} \sim 10^{-2} \Rightarrow |\Delta f_{\rm NL}^{\rm sq}| \sim 3 \times 10^{-3}$, very small.

Equilateral: $\delta_{\rm mem} \sim 10^{-2}$–$10^{-1}$, $\ln(k/k_{\rm nuc}) \sim 10 - 50$ gives $f_{\rm NL}^{\rm mem} \sim 0.1 - \text{few}$.

To conclude

- Introduced a causal memory kernel directly in the Wheeler-DeWitt equation, at the same m_P^{-2} order as standard QG corrections, preserving the classical background and Quantum Mechanics .
- Under a plus distribution kernel, this memory term admits an effective description as a Caputo fractional time derivative.
- The result: a new $k^{3/4}$ blue-tilted correction to the primordial power spectrum and temperature anisotropies, strongest at high multipoles ℓ, plus similar signatures in the bispectrum.
- Next generation experiments retain sensitivity to such memory effects.

Future Directions

- **Structure formation:** Enhanced small-scale temperature anisotropies can influence the abundance and character of bound structures, galaxies, stellar systems, planetary environments.
- **Primordial black holes:** The blue-tilted correction could shift PBH abundance in specific mass windows.

Thank you.

Questions and discussion welcome.

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